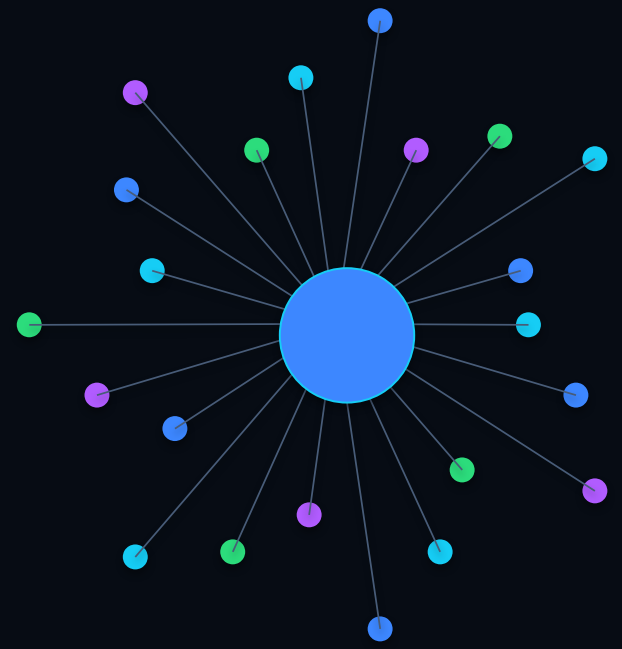
AI Learning Portfolio

AI Literacy • Generative AI • IBM Granite • Prompting • RAG • Customer Service • Multiagent Systems • Emerging Tech



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Goal: turn separate AI certificates into one connected, readable presentation.

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Learning Map

From basic AI understanding to advanced AI workflows



1 Foundation

First we build the basic vocabulary: what AI is, how data is used, and what AI can do.

2 Tools and workflows

Then we connect models, prompts, RAG, and customer service applications into practical use cases.

3 Advanced systems

Finally we look at AI agents, multiagent systems, emerging technologies, and responsible use.

Read the deck as one story: AI skills start with understanding, then grow into useful systems.

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AI Literacy: Understanding AI Before Using It

AI literacy means knowing what AI can do, where it can fail, and how to use it responsibly.

🔎 Recognize AI applications

AI appears in chatbots, recommendations, spam filters, translation tools, navigation, and facial recognition.

🧠 Understand AI systems

AI systems learn patterns from data and use those patterns to predict, generate, classify, or decide.

⚠️ Evaluate limits

Outputs may be wrong, biased, outdated, or incomplete, so human checking is still needed.

✓ Use responsibly

Fairness, privacy, transparency, and ethics must guide all AI use.

💬 Simple explanation

AI literacy is not about becoming a programmer immediately. It is about being able to ask: What is the system doing? What data is behind it? Can I trust this result? What could go wrong?

🎯 Why it matters

AI tools are now used in study, work, communication, security, and business. A person with AI literacy can use these tools faster while avoiding blind trust and careless data sharing.

Main idea: AI literacy = useful knowledge + critical thinking + responsible action.

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What AI Does With Data

AI turns raw data into predictions, generated content, classifications, and decisions.

▦ Structured data

Organized in tables, spreadsheets, and databases.
Example: customer ID, product, price, date.

{} Semi-structured data

Partly organized but flexible.
Example: emails, JSON files, XML, web logs.

◈ Unstructured data

No fixed table format.
Example: text, images, audio, video, PDFs.

CORE AI CAPABILITIES

A Language processing

Understand, summarize, translate, and generate human language.

👁 Perception

Interpret images, sound, and sensor data, such as face unlock or object detection.

📈 Prediction

Use patterns in past data to forecast likely future results or user behavior.

→ Decision AI

Choose a suitable action from available options.

AI is only as strong as the data, task definition, and human review around it.

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Generative AI and Large Language Models

Generative AI creates new content; LLMs are a major type that generate and understand language.

✦ Generative AI

Creates new text, images, code, music, or summaries by learning patterns from training data.

It is useful for drafting, brainstorming, and transforming information.

LLM Large Language Models

LLMs predict and generate language based on context.

They can answer questions, write emails, explain topics, classify text, and summarize documents.

! Important limitation

LLMs do not “know” in the human sense.

They can produce confident but wrong answers, so verification is necessary.

Input / prompt → **Model finds patterns** → **Generates output** → **Human checks result**

Best use: let AI support thinking and productivity, but keep people responsible for final quality.

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IBM Granite: Classification and Summarization Tasks

Granite models can support business tasks such as understanding feedback, classifying messages, and producing summaries.

🏷️ Classifying data

The model places input into labels such as positive / neutral / negative, urgent / medium / low, or product / payment / delivery.

≡ Summarizing data

The model condenses long reviews, reports, emails, or chat logs into short insights that are easier to read.

⚙️ Why businesses use it

It reduces manual workload, speeds up analysis, and helps teams react to customer issues faster.

Example Example: customer feedback

Raw review: “The delivery was late, but support solved the problem quickly.”

Classification: mixed / delivery issue.

Summary: customer had a delivery delay but positive support experience.

✓ Quality checklist

Use clear labels, give examples, check edge cases, and review model outputs. Classification is powerful, but ambiguous text still needs human judgment.

Granite-style AI tasks show a practical point: AI is useful when the task, labels, and evaluation are clear.

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Prompt Engineering: How to Ask Better Questions

A prompt is the instruction given to an AI model; better prompts usually produce better outputs.

Task

What should the AI do?

Context

What background information does it need?

Format

Should the answer be a table, paragraph, bullets, email, or code?

Tone

Should it sound formal, friendly, simple, technical, or persuasive?

Constraints

What limits should it follow, such as word count, audience, or excluded content?

↔ Weak vs. stronger prompt

Weak: “Explain AI.”

Stronger: “Explain AI to first-year students in 5 bullet points. Use simple language and include one everyday example.”

The stronger prompt gives the model a task, audience, format, and style.

🛠️ Prompt techniques

Zero-shot: ask directly without examples.

Few-shot: give examples first.

Templates: reuse a structure with variables.

Iteration: test, analyze, refine, and improve.

Prompting is not magic. It is clear communication with a system that follows patterns.

Retrieval-Augmented Generation (RAG)

RAG improves AI answers by adding relevant external information before the model responds.

Problem **Problem with normal LLMs**

They may answer from outdated training data or invent details when they lack information.

RAG **RAG solution**

Search a trusted knowledge source, retrieve relevant chunks, and provide them to the model as context.

Result **Result**

Answers become more grounded, more current, and easier to verify against the retrieved source.



Think of RAG as “AI with a library”: the model answers after reading the right material.

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RAG Components and Best Practices

A good RAG system depends on retrieval quality, not only on the language model.

1 Retriever

Finds the most relevant documents or chunks for the query.

2 Embedding model

Converts text into vectors so similar meanings can be matched.

3 Vector database

Stores embeddings and enables fast similarity search.

4 LLM generator

Uses the query plus retrieved context to create the final response.

✓ Keep knowledge updated

Outdated documents create outdated answers. A RAG system needs maintained and trusted sources.

✓ Chunk effectively

Split long documents into meaningful pieces. Chunks that are too large or too small reduce retrieval quality.

✓ Test and compare

Compare baseline LLM answers with RAG answers. Check accuracy, relevance, and missing context.

RAG reduces hallucinations, but does not remove the need for testing and human review.

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AI-Enabled Applications for Customer Service

AI helps support teams respond faster, route issues correctly, and learn from customer interactions.

💬 Chatbots and virtual assistants

Provide instant 24/7 answers for common questions and reduce repetitive work for human agents.

❤️ Sentiment analysis

Detects whether a message is positive, neutral, or negative and helps prioritize angry or urgent customers.

🔀 Ticket routing

Automatically sends a support request to the correct department or agent based on topic and urgency.

★ Personalization

AI can use customer history, preferences, and previous interactions to suggest relevant products or give tailored responses.

Human Human role

AI handles repetitive work, but humans remain important for empathy, complicated cases, complaints, and final decisions.

Good customer service AI combines speed with human trust, not automation alone.

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Example Workflow: AI Support Assistant

One practical system can combine classification, retrieval, generation, and analytics.

1. Customer message

“My order is late and no one replied.”

2. Classify urgency

Topic: delivery. Sentiment: negative. Priority: high.

3. Retrieve policy

RAG finds refund, delivery, and escalation rules.

4. Generate reply

AI drafts a helpful, policy-based response.

5. Human review

Agent checks tone and approves or edits.

6. Analyze trends

Managers see repeated delivery complaints.

This is the useful combination: AI handles routine structure; people handle judgment and relationship.

AI Agents: From Chat Responses to Actions

An AI agent perceives information, decides what to do, and takes actions toward a goal.

Perceive

Collect information from users, tools, sensors, databases, or the environment.

Reason / decide

Interpret the situation and choose a plan based on goals and constraints.

Act

Use tools, send messages, update systems, search data, or trigger workflows.

Learn / adapt

Use feedback and outcomes to improve future decisions.

1 Single-agent example

A chatbot that answers a customer question by itself using one model and one task flow.

2 Agentic workflow example

An assistant that reads a message, searches company policy, drafts a reply, creates a ticket, and alerts a human.

The key difference: an agent is not only generating text; it is working toward a goal.

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The Rise of Multiagent Systems

Multiagent systems use several autonomous agents that collaborate, communicate, or compete.

🤝 Collaboration

Agents share information and divide work to solve a common problem faster.

↔ Communication

Agents exchange messages to coordinate actions or negotiate decisions.

⚔️ Competition

Agents may have different goals and compete for limited resources or better outcomes.

Why Why MAS is rising

Modern problems involve large data, many systems, and changing environments. Multiple agents can divide tasks and work in parallel.

+ Benefits

Scalability, flexibility, robustness, parallel task execution, and better handling of real-world complexity.

Use Applications

Traffic control, financial trading, cybersecurity, supply chains, recommendations, and smart cities.

Multiagent systems are useful when one AI system alone is too limited for the whole problem.

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Emerging Technologies Around AI

AI grows stronger when it is connected with data, cloud, cybersecurity, and new computing ideas.

☁️ Cloud computing

Provides scalable storage and computing power so AI systems can run for many users and large datasets.

🛡️ Cybersecurity

Uses AI for threat detection, anomaly monitoring, and faster response, while also needing protection from misuse.

📊 Data science

Collects, cleans, analyzes, and visualizes data so AI models can learn from reliable information.

Q Quantum computing

Explores qubits, superposition, interference, measurement, and future ways of solving difficult problems.

⚡ Emerging tech credential

Connects AI, cloud, cybersecurity, data, and other digital technologies as part of a modern skill foundation.

ƒ Automation computability

Focuses on what machines can compute, automate, and decide through defined processes and algorithms.

Together, these topics show that AI is not isolated; it is part of a full digital ecosystem.

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Ethical Considerations for Generative AI

Responsible AI is the layer that keeps powerful systems safe, fair, and useful.

⚖️ Bias and fairness

Training data may contain social bias. AI outputs should be checked for unfair or discriminatory results.

🔒 Privacy and security

AI systems may process personal or sensitive data, so data protection and access control matter.

🔍 Transparency

Users should understand when AI is involved and how important decisions are supported.

✓ Accuracy checks

Generative AI can hallucinate. Claims, summaries, and recommendations should be verified against reliable sources.

Human Human oversight

Humans should monitor high-impact decisions and remain accountable for final outcomes.

Policy Safe use

Do not upload sensitive data unnecessarily. Use AI tools according to policy, law, and ethical expectations.

Ethics is not an extra topic at the end; it must be part of every AI workflow.

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Combined AI Workflow: How the Topics Connect

A real AI solution usually combines many of the skills from this portfolio.



Project **Example project idea**

An AI support assistant receives customer messages, classifies urgency, retrieves company policy with RAG, drafts a reply, routes complex cases to the correct team, and summarizes feedback trends for managers.

Value **Why this is strong**

The system combines AI literacy, prompt engineering, reliable retrieval, customer service automation, agentic action, and responsible oversight in one practical workflow.

The portfolio is not just many certificates; it forms a practical toolkit for building AI-enabled solutions.

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Completed SkillsBuild Certificate Portfolio

The presentation combines the main learning themes from the completed certificates shown by the user.

- Introduction to Artificial Intelligence
- Exploring Artificial Intelligence
- Interacting with AI
- Foundations in Generative AI
- Introduction to Large Language Models
- Ethical Considerations for Generative AI
- Classifying Data Using IBM Granite
- Summarizing Data Using IBM Granite
- Craft Precise Prompts for AI Models
- Retrieval-Augmented Generation
- AI-Enabled Applications for Customer Service
- The Rise of Multiagent Systems
- Exploring Data
- Exploring Cloud Computing
- Exploring Cybersecurity
- Exploring Quantum Computing
- Emerging Tech Digital Credential
- Automata Computability

This slide can be used as a quick proof-of-learning overview during a presentation.

Career Value of the Combined Topics

Together, these skills support both technical awareness and practical AI use.

1 Technical awareness

Understand AI, data, cloud, cybersecurity, quantum concepts, and modern computing trends.

2 Practical AI use

Use AI for writing, summarizing, classifying, support automation, and better decision-making.

3 Responsible mindset

Recognize bias, privacy, hallucination, transparency, and human oversight requirements.

Build Project readiness

The topics are useful for building prototypes such as AI support assistants, prompt-refinement tools, feedback analyzers, and RAG knowledge bots.

Present Presentation strength

Because the slides contain explanations, they can be read directly during presenting while still looking clean and professional.

Result: a broad AI foundation with both conceptual understanding and application areas.

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Conclusion

AI becomes valuable when knowledge, tools, data, and responsibility work together.

AI skills are strongest when combined:

- understanding
- prompting
- data and retrieval
- automation
- human responsibility

🎤 Final message to say

This portfolio shows a complete learning path: from basic AI literacy to generative AI, prompts, RAG, customer service applications, agents, multiagent systems, and responsible AI use.

Key takeaway: AI is a practical future skill, but trustworthy AI always needs good data, clear goals, and human judgment.

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Source Material Used

Summarized and redesigned from user-provided IBM SkillsBuild certificate screenshots and presentation PDFs.

Introduction to Artificial Intelligence / AI Literacy: AI basics, capabilities, benefits, limits, and responsible use.

AI-Enabled Applications for Customer Service: chatbots, sentiment analysis, personalization, ticket routing, and challenges.

Craft Precise Prompts for AI Models: task, context, format, tone, constraints, templates, and iteration.

Retrieval-Augmented Generation: workflow, components, use cases, challenges, and best practices.

The Rise of Multiagent Systems: agents, collaboration, communication, competition, benefits, and applications.

Additional certificates shown in screenshots: Generative AI, LLMs, IBM Granite, data, cloud, cybersecurity, quantum, and emerging technologies.

This version is designed to be readable from the slides during presentation.